Physics Enhanced Deep Surrogates for the one-dimensional diffusion equation

October 31, 2024

1 Problem specification

Consider the unit interval $\Omega = [0, 1]$. Let the function $f : \Omega \to \mathbb{R}$ be fixed. For a given function $\alpha : \Omega \to \mathbb{R}$, we want to find the function $u : \Omega \to \mathbb{R}$ which satisfies

$$-\frac{d}{dx}\left(\exp[\alpha(x)]\frac{du}{dx}(x)\right) = f(x) \qquad \text{for all } 0 < x < 1 \tag{1}$$

subject to the boundary conditions

$$\begin{aligned} u(x = 0) &= 0 \qquad \text{(homogeneous Dirichlet)} \\ \frac{du}{dx}(x = 1) &= -g \exp[-\alpha(1)] \qquad \text{(von Neumann)} \end{aligned} \tag{2}$$

In a more abstract sense, let V be the space of real-valued functions defined on Ω. We want to find the map $\mathcal{F} : V \to V$ with $\mathcal{F} : \alpha \mapsto u(\alpha)$ where $u = u(\alpha)$ is defined by (1) and (2).

Physical interpretation Note that by setting $K(x) := \exp[\alpha(x)]$, (1) can be written as $-\nabla \cdot (K\nabla u) = f$. So physically the problem is about finding the solution of the diffusion equation for a given spatially varying diffusion coefficient $K(x) > 0$ which is parametrised by its logarithm $\alpha(x)$.

Imagine, for example, that u is a temperature field. Then the heat flux is given by

$$\phi = -K\nabla u \tag{3}$$

where K is the spatially varying thermal conductivity. If f describes the sources and sinks of heat, then the (stationary) conservation law is

$$\nabla \cdot \phi = f. \tag{4}$$

The boundary conditions in (2) correspond to fixing the temperature at $x = 0$ and fixing the heat flux at $x = 1$.

1.1 Quantity of interest

Usually we are not really interested in predicting $u(x)$ itself for a given $\alpha(x)$ but instead want to compute some functional Q of $u(x)$

$$\mathcal{Q} : V \to \mathbb{R}^\beta. \tag{5}$$

In other words, we are interested in the map $\mathcal{Q} \circ \mathcal{F} : V \to \mathbb{R}^\beta$. $\mathcal{Q}$ is also called the "Quantity of Interest" (short QoI). Some examples are:

- Average field ($\beta = 1$):

$$\mathcal{Q}(u) = \int_\Omega u(x)\, dx \tag{6}$$

- Sampled field:

$$\mathcal{Q}(u) = (u(\xi_1), u(\xi_2), \ldots, u(\xi_\beta)) \tag{7}$$

for some sample points $\xi_1, \xi_2, \ldots, \xi_\beta \in \Omega$.

- Higher moments of field:

$$\mathcal{Q}(u) = \left(\int_\Omega u(x)^{\nu_1}\, dx, \int_\Omega u(x)^{\nu_2}\, dx, \ldots, \int_\Omega u(x)^{\nu_\beta}\, dx \right) \tag{8}$$

for some powers $\nu_1, \nu_2, \ldots, \nu_\beta \in \mathbb{N}$.

Note that $\mathcal{Q}$ is linear in the first two cases but non-linear in the last case.

2 Discretisation

Let $\Omega = [0,1]$ be the unit interval. Split Ω into n sub-intervals of size $h = 1/n$ to obtain a grid $\Omega_h := \bigcup_{j=0}^{n-1} I_j^{(h)}$ where $I_j^{(h)} = [jh, (j+1)h] = [x_j^{(h)}, x_{j+1}^{(h)}]$ for $j = 0, 1, \ldots, n-1$ is the j-th sub-interval or cell of the grid. We call the points $x_j^{(h)} = jh$ for $j = 0, 1, \ldots, n$ the vertices of the grid.

Let $\alpha^{(h)} \in \mathbb{R}^{n+1}$ be the vector with $\alpha_j^{(h)} \approx \alpha(x_j^{(h)})$ for $j = 0, 1, \ldots, n$ and $u^{(h)} \in \mathbb{R}^n$ be the vector with

$$u_j^{(h)} \approx \frac{1}{h} \int_{I_j^{(h)}} u(x)\, dx \qquad \text{for } j = 0, 1, \ldots, n-1, \tag{9}$$

with an analogous definition of the vector $f^{(h)} \in \mathbb{R}^n$. The definition in (9) says that the components of the vector $u^{(h)}$ approximate the volume averages of the solution in the grid cells. To discretise (1) with the finite volume method, integrate the equation over cell $I_j^{(h)}$ and use the boundary conditions in (2):

$$\begin{aligned} h f_j^{(h)} := \int_{I_j^{(h)}} f(x)\, dx &= -\int_{I_j^{(h)}} \frac{d}{dx}\left(\exp[\alpha(x)] \frac{du}{dx}(x) \right)\, dx \\ &= \exp[\alpha(x_j^{(h)})] \frac{du}{dx}(x_j^{(h)}) - \exp[\alpha(x_{j+1}^{(h)})] \frac{du}{dx}(x_{j+1}^{(h)}) \\ &\approx \begin{cases} \exp[\alpha_j^{(h)}] \frac{2u_j^{(h)}}{h} - \exp[\alpha_{j+1}^{(h)}] \frac{u_{j+1}^{(h)} - u_j^{(h)}}{h} & \text{for } j = 0 \\ \exp[\alpha_j^{(h)}] \frac{u_j^{(h)} - u_{j-1}^{(h)}}{h} - \exp[\alpha_{j+1}^{(h)}] \frac{u_{j+1}^{(h)} - u_j^{(h)}}{h} & \text{for } 0 < j < n-1 \\ \exp[\alpha_j^{(h)}] \frac{u_j^{(h)} - u_{j-1}^{(h)}}{h} + g & \text{for } j = n-1 \end{cases} \end{aligned} \tag{10}$$

This results in the following linear system:

$$A^{(h)} u^{(h)} = \widetilde{f}^{(h)} := f^{(h)} - \frac{g}{h} \widehat{e}^{(h)} \tag{11}$$

where $\widehat{e}^{(h)} = (0, 0, \ldots, 0, 1) \in \mathbb{R}^n$ is the vector which is zero everywhere except for its final entry. The matrix $A^{(h)} = A^{(h)}(\alpha^{(h)})$ is given by

$$A^{(h)}(\alpha^{(h)}) = \frac{1}{h^2} \begin{pmatrix} 2K_0 + K_1 & -K_1 & 0 & 0 & 0 & \cdots & 0 \\ -K_1 & K_1 + K_2 & -K_2 & 0 & 0 & \cdots & 0 \\ 0 & -K_2 & K_2 + K_3 & -K_3 & 0 & \cdots & 0 \\ 0 & 0 & -K_3 & \ddots & & & \\ 0 & 0 & 0 & & & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & -K_{n-2} & 0 \\ & & & & -K_{n-2} & K_{n-2} + K_{n-1} & -K_{n-1} \\ 0 & 0 & 0 & \cdots & 0 & -K_{n-1} & K_{n-1} \end{pmatrix} \tag{12}$$

where, for simplicity, we defined the diffusion coefficients as

$$K_j^{(h)} = K_j^{(h)}(\alpha_j^{(h)}) := \exp[\alpha_j^{(h)}]. \tag{13}$$

Note that by solving (11) for $u^{(h)}$ for a given $\alpha^{(h)}$ we can construct an approximation $\mathcal{F}_h$ of the map $\mathcal{F}_h$:

$$\mathcal{F}_h : \mathbb{R}^n \to \mathbb{R}^n, \qquad \alpha^{(h)} \mapsto u^{(h)} = u^{(h)}(\alpha^{(h)}) = \text{ the solution of (11) for given } \alpha^{(h)}. \tag{14}$$

We also need to construct an approximation $Q^{(h)}$ of the QoI in (5). For example, if the QoI is the average field, then $Q^{(h)} : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$Q^{(h)}(u^{(h)}) = \frac{1}{n}\sum_{j=0}^{n-1} u_j^{(h)}. \tag{15}$$

If the QoI is the field sampled at some locations $\xi_1, \xi_2, \ldots, \xi_q$, then $Q^{(h)} : \mathbb{R}^n \to \mathbb{R}^\beta$ is defined by

$$Q^{(h)}(u^{(h)}) = \left(u_{j_1}^{(h)}, u_{j_2}^{(h)}, \ldots, u_{j_\beta}^{(h)}\right) \qquad \text{with } \xi_k \in I_{j_k}^{(h)} \text{ for } k = 1, 2, \ldots, \beta. \tag{16}$$

3 Solution procedure

3.1 Pure Numerical solution

1. Given $\alpha^{(h)} \in \mathbb{R}^{n+1}$, compute $A^{(h)} = A^{(h)}(\alpha^{(h)})$.
2. Solve the linear system in (11) to obtain $u^{(h)} = u^{(h)}(\alpha^{(h)}) \in \mathbb{R}^n$.
3. Compute the quantity of interest $q^{(h)} = q^{(h)}(\alpha^{(h)}) = Q^{(h)}(u^{(h)}(\alpha^{(h)})) \in \mathbb{R}^\beta$

This method can be made very accurate by choosing $h \ll 1$, but it also becomes very expensive in this case. The computational bottleneck is the solution of (11).

3.2 Pure ML solution

1. Generate a training data set $\{(\alpha^{(h)\ell}, q^{(h)\ell})\}_{\ell=1}^{N_{\text{samples}}}$ where for each pair $(\alpha^{(h)\ell}, q^{(h)\ell})$ we have that $q^{(h)\ell} = Q^{(h)}(u^{(h)\ell})$ and $u^{(h)\ell}$ satisfies (11), i.e.

$$A^{(h)}(\alpha^{(h)\ell})u^{(h)\ell} = \widetilde{f}^{(h)} \qquad \text{for } \ell = 1, 2, \ldots, N_{\text{samples}}. \tag{17}$$

2. Construct a neural network Φ_{NN} which is parametrised by some parameters θ

$$\Phi_{\text{NN}}(\cdot|\theta) : \mathbb{R}^{n+1} \to \mathbb{R}^\beta \tag{18}$$

3. Train the neural network by minimising some loss function $L(\{(\alpha^{(h)\ell}, q^{(h)\ell})\}_{\ell=1}^{N_{\text{samples}}}|\theta)$. For example, one could minimise the average (squared) L_2 error:

$$\begin{aligned} L(\{(\alpha^{(h)\ell}, q^{(h)\ell})\}_{\ell=1}^{M}|\theta) &= \frac{1}{N_{\text{samples}}}\sum_{\ell=1}^{N_{\text{samples}}} L_2(\alpha^{(h)\ell}, q^{(h)\ell}|\theta) \\ L_2(\alpha^{(h)}, q^{(h)}|\theta) &:= \left(\Phi_{\text{NN}}(\alpha^{(h)}|\theta) - q^{(h)}\right)^2 \end{aligned} \tag{19}$$

If trained properly, Φ_{NN} with optimal parameters θ_{opt} will give us a good approximation of the quantity of interest for any $\alpha^{(h)} \in \mathbb{R}^n$:

$$\Phi_{\text{NN}}(\alpha^{(h)}|\theta_{\text{opt}}) \approx q^{(h)} \tag{20}$$

where $q^{(h)} = Q^{(h)}(u^{(h)})$ and $u^{(h)}$ is the solution of $A^{(h)}(\alpha^{(h)})u^{(h)} = \widetilde{f}^{(h)}$. The problem with this approach is that the neural network is a black box.

3.3 Hybrid approach: Physics Enhanced Deep Surrogates (PEDS)

The following approach is described in [1]:

1. Generate a training data set $\{(\alpha^{(h)\ell}, q^{(h)\ell})\}_{\ell=1}^{N_{\text{samples}}}$ as in Step 1 of the Pure ML approach in Section 3.2.

2. Construct a second grid Ω_H with grid spacing $H = 1/N$ consisting of $N < n$ cells. We usually want to choose $H \gg h$.

3. Define the downsampling operator
$$R : \mathbb{R}^{n+1} \to \mathbb{R}^{N+1}, \qquad \alpha^{(h)} \mapsto \alpha_{\text{DS}}^{(H)} \tag{21}$$
which takes a high-resolution $\alpha^{(h)}$ and turns it into a low-fidelity downsampled $\alpha_{\text{DS}}^{(H)}$. If $H = \mu h$ where the scale parameter $\mu = H/h = n/N \in \mathbb{N}$ is an integer, then we can for example simply set
$$(\alpha_{\text{DS}}^{(H)})_j = \alpha_{\mu \cdot j}^{(h)} \qquad \text{for } j = 0, 1, \ldots, N. \tag{22}$$

4. Construct a neural network Ψ_{NN} parametrised by some parameters θ (which have nothing to do with the NN parameters in Section 3.2)
$$\Psi_{\text{NN}}(\cdot|\theta) : \mathbb{R}^n \to \mathbb{R}^N. \tag{23}$$
Ψ_{NN} maps $\alpha^{(h)}$ to some generated $\alpha_{\text{NN}}^{(H)}$ on the low-resolution grid.

5. Define some function $A : \mathbb{R}^{N+1} \times \mathbb{R}^{N+1} \times [0, 1] \to \mathbb{R}^{N+1}$ for combining the downsampled $\alpha_{\text{DS}}^{(H)}$ and the generated $\alpha_{\text{NN}}^{(H)}$ into some $\alpha^{(H)} = A(\alpha_{\text{DS}}^{(H)}, \alpha_{\text{NN}}^{(H)}, w)$ where w is a learnable parameter. For example, one might choose one of the following two weighted means
$$\alpha_j^{(H)} = w(\alpha_{\text{NN}}^{(H)})_j + (1 - w)(\alpha_{\text{DS}}^{(H)})_j \qquad \text{for } j = 0, 1, \ldots, N \tag{24}$$
or
$$\alpha_j^{(H)} = \log\left(w \exp[(\alpha_{\text{NN}}^{(H)})_j] + (1 - w) \exp[(\alpha_{\text{DS}}^{(H)})_j]\right) \qquad \text{for } j = 0, 1, \ldots, N \tag{25}$$

6. For given neural network parameters θ and weight w define the function $\mathcal{F}_{\text{PEDS}}^{(h)}(\cdot|\theta, w) : \mathbb{R}^{n+1} \to \mathbb{R}^{N+1}$. This function maps a high-resolution $\alpha^{(h)}$ to the low fidelity solution $u^{(H)} = u^{(H)}(\alpha^{(h)})$ as follows:
   - Compute $\alpha_{\text{DS}}^{(H)}$ and $\alpha_{\text{NN}}^{(H)}$ and combine them into $\alpha^{(H)}$ as described above.
   - Compute the low-fidelity solution $u^{(H)} \in \mathbb{R}^N$ by solving $A^{(H)}(\alpha^{(H)})u^{(H)} = \widetilde{f}^{(H)}$

7. Find optimal θ_{opt} and w_{opt} by minimising some loss function $L(\{(\alpha^{(h)\ell}, q^{(h)\ell})\}_{\ell=1}^{N_{\text{samples}}}|\theta)$. Again, the natural choice is to minimise the average (squared) L_2 error:
$$\begin{aligned} L(\{(\alpha^{(h)\ell}, q^{(h)\ell})\}_{\ell=1}^{M}|\theta) &= \frac{1}{N_{\text{samples}}} \sum_{\ell=1}^{N_{\text{samples}}} L_2(\alpha^{(h)\ell}, q^{(h)\ell}|\theta) \\ L_2(\alpha^{(h)}, q^{(h)}|\theta) &:= \left(Q^{(H)}\left(\mathcal{F}_{\text{PEDS}}(\alpha^{(h)}|\theta, w)\right) - q^{(h)}\right)^2 \end{aligned} \tag{26}$$

If trained properly, $\mathcal{F}_{\text{PEDS}}$ with optimal parameters θ_{opt}, w_{opt} will give us a good approximation of the quantity of interest for any $\alpha^{(h)} \in \mathbb{R}^n$:
$$Q^{(H)}\left(\mathcal{F}_{\text{PEDS}}^{(h)}(\alpha^{(h)}|\theta_{\text{opt}}, w_{\text{opt}})\right) \approx q^{(h)} \tag{27}$$
where $q^{(h)} = Q^{(h)}(u^{(h)})$ and $u^{(h)}$ is the solution of $A^{(h)}(\alpha^{(h)})u^{(h)} = \widetilde{f}^{(h)}$.

A Gradients

We need to back-propagate through the solution, i.e. the map $\alpha \mapsto u$ defined by

$$\begin{aligned} A(\alpha)u &= \widetilde{f} \\ \Leftrightarrow \qquad u &= A(\alpha)^{-1}\widetilde{f} \end{aligned} \tag{28}$$

where for simplicity we have dropped all superscripts relating to the grid spacing H. The chain rule gives:

$$\frac{\partial L}{\partial \alpha_i} = \sum_j \frac{\partial u_j}{\partial \alpha_i}\frac{\partial L}{\partial u_j} \tag{29}$$

which leads to

$$\begin{aligned} \frac{\partial u_j}{\partial \alpha_i} &= \sum_k \left(\frac{\partial}{\partial \alpha_i} A(\alpha)^{-1}\right)_{jk} \widetilde{f}_k \\ &= -\left(A(\alpha)^{-1} D_i(\alpha) A(\alpha)^{-1}\widetilde{f}\right)_j \qquad \text{with } D_i(\alpha) := \frac{\partial \delta}{\partial \alpha_i}(\alpha). \end{aligned} \tag{30}$$

The matrix $D_i(\alpha)$ contains mostly zero entries. For $i = 0$ it has exactly one non-zero entry, namely $(D_0(\alpha))_{0,0} = 2h^{-2}\exp[\alpha_0]$. For $1 < i < N$ it has exactly four non-zero entries namely $(D_i(\alpha))_{i,i} = (D_i(\alpha))_{i-1,i-1} = -(D_i(\alpha))_{i,i-1} = -(D_i(\alpha))_{i-1,i} = h^{-2}\exp[\alpha_i]$.

Further, we can write:

$$\frac{\partial L}{\partial \alpha_i} = -w^\top D_i(\alpha) u \tag{31}$$

with

$$A(\alpha)w = \frac{\partial L}{\partial u}, \qquad\qquad A(\alpha)u = \widetilde{f} \tag{32}$$

B Distribution of diffusion coefficient

We assume that the logarithm $\alpha(x)$ of the diffusion coefficient is a Gaussian Random Field (GRF) which satisfies

$$(-\Delta + \kappa^2)^{a/2}\alpha(x) = \mathcal{W}(x) \tag{33}$$

where $\mathcal{W}$ is white noise with $\mathbb{E}[\mathcal{W}(x)] = 0$, $\mathbb{E}[\mathcal{W}(x)\mathcal{W}(y)] = \delta(x-y)$. $\kappa^{-1} =: \Lambda$ is the correlation length. We have trivially that $\mathbb{E}[\alpha(x)] = 0$. If the field was defined on the entire real line then its covariance would be given by the Matern function

$$\mathrm{Cov}[\alpha(x)\alpha(y)] = \mathbb{E}[\alpha(x)\alpha(y)] = C_{\nu,\kappa}(\kappa|x-y|) := \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}\left(\kappa|x-y|\right)^\nu K_\nu(\kappa|x-y|) \tag{34}$$

where $a = \nu + \frac{1}{2}$ and K_ν is the modified Bessel function of the second kind [2]. The marginal covariance is given by

$$\sigma^2 = \sigma^2(\nu,\kappa) = \frac{\Gamma(\nu)}{\Gamma(\nu+\frac{1}{2})\sqrt{4\pi}\kappa^{2\nu}} \tag{35}$$

Here we only consider the two cases $a = 1$, $\nu = \frac{1}{2}$ and $a = 2$, $\nu = \frac{3}{2}$ for which the Matern function $C_{\nu,\kappa}$ reduces to

$$\begin{aligned} C_{\nu=\frac{1}{2},\kappa}(\kappa|x-y|) &= \frac{1}{2\kappa}\exp[-\kappa|x-y|], \\ C_{\nu=\frac{3}{2},\kappa}(\kappa|x-y|) &= \frac{1}{4\kappa^3}\left(1+\kappa|x-y|\right)\exp[-\kappa|x-y|]. \end{aligned} \tag{36}$$

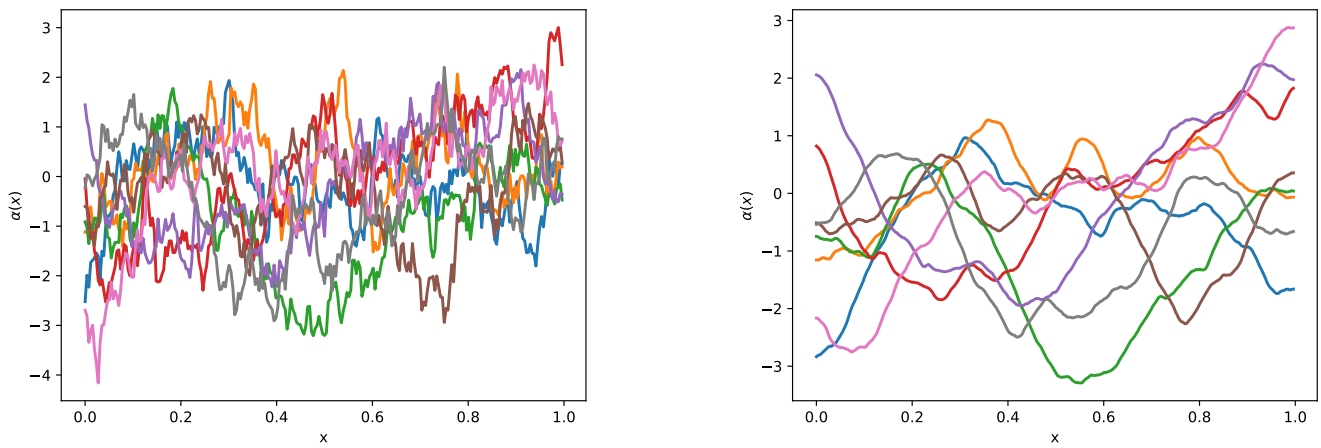

Figure 1: Random samples $\alpha(x)$ drawn from $\pi^{(h)}(\cdot|\nu,\kappa)$ for $a=1$, $\nu=\frac{1}{2}$ (left) and $a=2$, $\nu=\frac{3}{2}$ (right). In both cases the correlation length is $\kappa^{-1}=0.1$ and the number of gridcells is $N=256$. The samples are scaled by σ^{-1} to make their variance ~ 1.

For the finite domain $\Omega=[0,1]$ we need to impose boundary conditions which we choose to be $\frac{d\alpha}{dx}(0)=\frac{d\alpha}{dx}(1)=0$. To generate samples of $\alpha(x)$ in practice, we divide the domain into N cells as above and sample the vector $\alpha^{(h)}=(\alpha(x_0),\alpha(x_1),\ldots,\alpha(x_N))\in\mathbb{R}^{N+1}$ at the vertices x_j. A finite difference discretisation of the shifted Laplace operator $-\Delta+\kappa^2$ is given by the tridiagonal matrix

$$Q^{(h)}=\frac{1}{h^2}\begin{pmatrix} 2+\kappa^2h^2 & -1 & 0 & 0 & 0 & \cdots & 0 \\ -1 & 2+\kappa^2h^2 & -1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 2+\kappa^2h^2 & -1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & \ddots & & & \\ 0 & 0 & 0 & & & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & -1 & 0 \\ & & & & -1 & 2+\kappa^2h^2 & -1 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 2+\kappa^2h^2 \end{pmatrix}. \tag{37}$$

Let $\xi^{(h)}=(\xi_0,\xi_1,\ldots,\xi_N)\in\mathbb{R}^{N+1}$ be a vector of independent and identically distributed (i.i.d.) Gaussian random variables $\xi_j\sim\mathcal{N}(0,h^{-1})$ for $j=0,1,\ldots,N$; we also write $\xi^{(h)}\sim\pi_0^{(h)}$. For $a=1$, $\nu=\frac{3}{2}$ we can generate a sample $\alpha^{(h)k}$ from the target distribution $\pi^{(h)}(\cdot|\nu,\kappa)$ by drawing $\xi^{(h)k}\sim\pi_0^{(h)}$ and solving the tridiagonal system $Q^{(h)}\alpha^{(h)k}=\xi^{(h)k}$ for $\alpha^{(h)k}$. For $a=1$, $\nu=\frac{1}{2}$ we Cholesky factorise $Q^{(h)}=L^{(h)}(L^{(h)})^\top$ and solve the bi-diagonal system $(L^{(h)})^\top\alpha^{(h)k}=\xi^{(h)k}$. Observe that the approach is readily extended to arbitrary integer $a>1$. Fig. 1 showns some random samples generates in this way for both $a=1$, $\nu=\frac{1}{2}$ and $a=2$, $\nu=\frac{3}{2}$. The corresponding covariance between a and $y=\frac{3}{4}$ is visualised in Fig. 2 where we show both an estimator for the covariance on the interval $\Omega=[0,1]$ and the Matern covariance function. The two functins differ since von Neumann boundary conditions have been imposed to generate the samples.

References

[1] Raphaël Pestourie, Youssef Mroueh, Chris Rackauckas, Payel Das, and Steven G Johnson. Physics-enhanced deep surrogates for partial differential equations. *Nature Machine Intelligence*, 5(12):1458–1465, 2023.

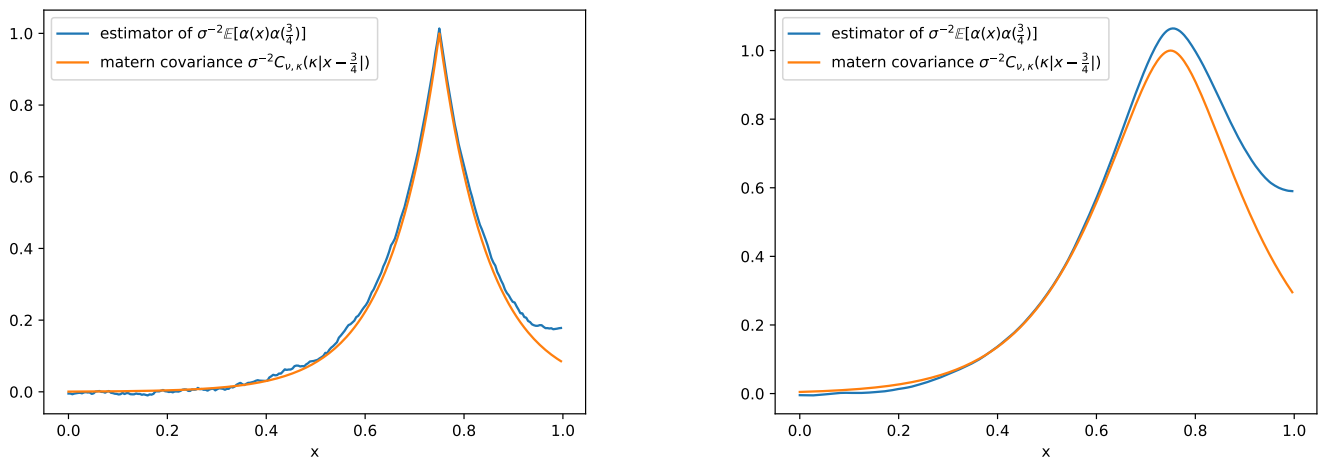

Figure 2: Empirical covariance between the points x and $y = \frac{3}{4}$ and Matern covariance function. Results are shown for $a = 1$, $\nu = \frac{1}{2}$ (left) and $a = 2$, $\nu = \frac{3}{2}$ (right). In both cases the correlation length is $\kappa^{-1} = 0.1$ and the number of gridcells is $N = 256$. The samples are scaled by σ^{-1} to make their variance ~ 1.

[2] Finn Lindgren, Håvard Rue, and Johan Lindström. An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential equation approach. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 73(4):423–498, 2011.